



Organic Residues

Several classes of organic molecules can be analyzed, although the techniques needed to identify each vary. Before we cover those techniques, let's review the structure of the organic molecules most often preserved archaeologically and discuss how, when, and where they are preserved well or poorly.

Proteins

Proteins are **polymers** of **amino acids**, 20 of which exist. Proteins can consist of not only a single chain of amino acids (termed a polypeptide) but also a complex, three-dimensional structure of multiple polypeptide strands. One common way that polypeptides are depicted is as a chain of amino acids, each abbreviated with a single letter. Another way is as a three-dimensional structure that shows the physical form of the protein (including multiple polypeptide strands, when present), as inferred from its structure and study.

1	MEPRVVKPPGQDLVVESLKSRVGLGGSCPDEYDFSIFYQSKYKRRLTSP	50
51	GDLDIYSGDKVGSSSLKYSDESKHCRTPLGSLFKHVNVCNCLDDELDSEFHD	100
101	KKQETEEELIENDYRVSTSKITKQSFKEIEKVALPTNTTSSRPRTCCSD	150
151	AGDSPLKPVSCPKSKASDKRSLPHQISQIYDELFIHLKLQCETAAQK	200
201	FAEELQKRERFLLEREQLLFRHENALSKIKGVVEEVLTFRFQIIKEQHDAE	250
251	VEHLETVLKEKNKETKRLRSSFDALKELNDTLKKQLNEASEENRKIDIQA	300
301	KRVQARLDNLQRKYEFMTIQRLKGSSHAVHEMKSLEKAPVSKTYKVPL	350
351	NGQVYELLTVFMDWISDHLSKVKHEESGMDGKKPQLKFASQRNDIQEKC	400
401	VKLLPLMTEQLQWMPFVNKLHEPFVKFIYWSLRQLDAGAHSTMTSTLR	450
451	RLGEDIFKGVVTKGIQDNSPQHSVENKPKTAFFKSSNPLRFLSTLIVL	500
501	KTVTQADYLAQAFDSLCLDLKTEEGKTLFLEYQAVPVILSHLRISKGLL	550
551	SNVIDSLLQMTVESKSLQPFLEACSNLFFRTCSVLLRAPKLDLQILEKL	600
601	SIIQLKLSKIKSNKKLFELFTIHLMLQEIQRTTNPEHAFLCINLSTLFN	650
651	LGLTKCNLSLVSSASP	700

An amino acid-sequence diagram for a protein known as CCDC138.

Source: [Wikimedia Commons](#)

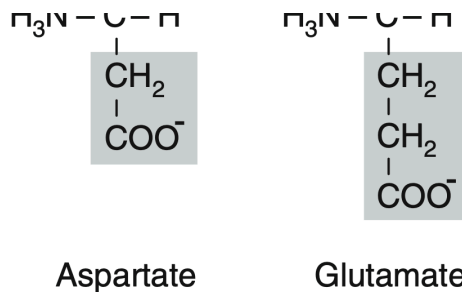
(https://commons.wikimedia.org/wiki/File:CCDC138_primary_protein_sequence.png)



Polymers are molecules formed of connected subunits, each termed a monomer. Examples include proteins (chains of amino acids) and carbohydrates (chains of sugars).

Amino acids are monomers that form proteins. Each shares a common structure with an amino group that holds a positive charge and an acid group with a negative charge, and these link to form the “backbone” of the protein polymer. Amino acids differ in the so-called “R-group,” which comes out from the side of the polypeptide chain.





Two amino acids, aspartate and glutamate. The area marked in grey is the R-group. The amino group (H_3N^+) and acid group (COO^-) can be seen on each.

Source: Malainey, M. E. (2010). Consumer's guide to archaeological science: Analytical techniques (https://bu.primo.exlibrisgroup.com/discovery/fulldisplay?vid=01BOSU_INST:BU&tab=Everything&docid=alma99207906539201161&searchScope=default_scope&context=L&lang=en). Springer.

Proteins have a multitude of functions in living organisms. You may be familiar with the “spike” protein of the SARS-CoV-2 virus (which causes the disease COVID-19), which is the portion of the virus that binds to a host cell and allows the virus to infect that cell with its genetic material. Other well-known proteins include hemoglobin (which holds oxygen in blood cells); gluten (a structural protein in certain plant seeds, especially wheat); keratin (the protein that is the primary structural component of mammal hair and nails); casein (a common protein in milk, found in the “curd” fraction); and collagen (the structural protein in bone that gives bone elasticity, also found in cartilage). Many of these proteins are present in food, whether derived from plants or animals.

The three-dimensional structure of proteins, however, can be maintained only under certain conditions. Heat, acidic or basic conditions, and other environmental disturbances can degrade proteins. As a result, proteins are often poorly preserved in archaeological contexts, with the exception of collagen, which can be well preserved for long periods of time within bones. The amino acid sequences of some proteins can be highly distinctive in individual species. Proteins used to be matched using antibodies tailored to specific organisms, although a newer technology, described below, allows the use of special mass spectrometry methods to create “fingerprints” for proteins belonging to specific species, allowing rapid and accurate identification of species from the proteins present in a sample.

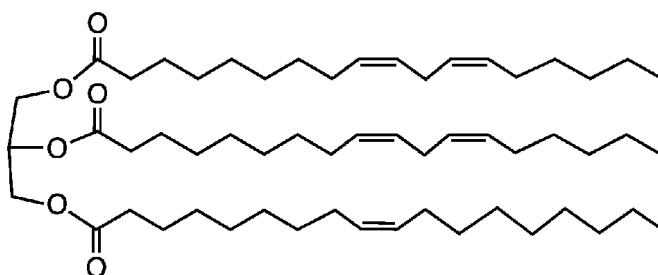
The archaeological context in which proteins are most likely to be preserved is bones. Naturally mummified or frozen plants or animals may also have excellent protein preservation, though such conditions are rare. Proteins can sometimes also be recovered from the edges of cutting tools, especially stone tools, when trapped and protected in tiny cracks or grooves in a tool's edge.

Lipids

Lipids are organic compounds defined by their low solubility in water, but solubility in nonpolar solvents (such as turpentine). There are many types of lipids, including waxes and sterols, but most common in animal and plants are **triglycerides**, molecules composed of one triacylglycerol backbone and three **fatty acid** chains, which may be the same or different from one another in a single triglyceride. Fats can be saturated—meaning that all carbon atoms in the fatty acid chains are connected by single covalent bonds—or unsaturated, meaning that one or more pair of carbon atoms are connected with a double bond. Double bonds create bends or "elbows" in the fatty acid chain, locations where it is easier for enzymes to break the chain, making unsaturated fatty acids easier to metabolize; hence, why unsaturated fats are "better" for you. Of course, there is more to the value of lipids as a source of nutrition than just how easy they are to metabolize! Saturated fatty acids also raise the melting point of a lipid, while unsaturated fatty acids lower it. This gives rise to the distinction between fats (e.g., butter) and oils (e.g., olive oil). Fats are lipids that are solids at room temperature, while oils are liquids. Of course, these common terms have exceptions; coconut oil is a solid at cool room temperatures (though will liquify in your kitchen on a very hot day!). You should thus not be surprised to learn that coconut oil has a greater proportion of saturated fatty acids than olive oil!

Triglycerides are a category of lipids that are the most common component of vegetable fats and adipose tissue (body fat) in animals. They are composed of one triacylglycerol backbone and three fatty acid chains.

Fatty acids are lipids and monomers of triglycerides. They contain a carboxylic acid group, which binds to the triacylglycerol in a triglyceride, and a chain of carbons with attached hydrogens up to 35 carbon atoms long.

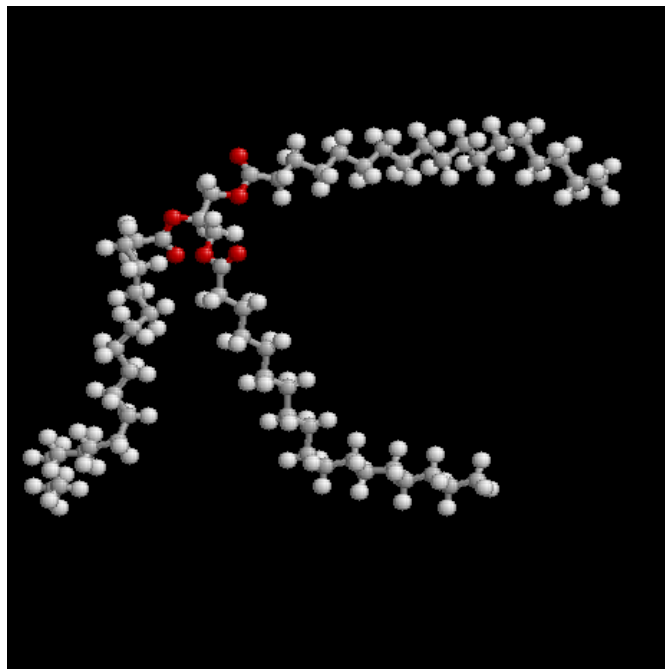


A triglyceride, sunflower oil. The triacylglycerol backbone is rendered on the left; the carbon chains of the fatty acids extend to the right. The

top two fatty acids are linoleic acid, and the bottom fatty acid is oleic acid. All are unsaturated, as indicated by the double line (representing a double bond between carbon atoms).

Source: [Wikimedia Commons](#)

(<https://commons.wikimedia.org/wiki/File:TriglycerideSunflower.png>)



A triglyceride rendered in three dimensions. The “elbows” in the fatty-acid chains are the locations of double bonds.

Source: [Wikimedia Commons](#)

(https://commons.wikimedia.org/wiki/File:Triglyceride_ballandstick.gif)

[GIF](#)

Lipids serve a number of functions in living organisms. A variety of special lipids formed from fatty acids are the primary component of cell membranes in all organisms. Triglycerides are primarily energy-storage molecules used by animals and plants. This is the primary reason why many seeds (botanically speaking, so including those we term nuts, such as almonds and cashews) have high lipid contents and are pressed for oil; the lipids are there as energy for the new plant that would ordinarily grow from the seed. Waxes have a variety of protective functions. Plants use them to reduce water loss in leaves, while birds use them to repel water from their feathers. Sterols (e.g., cholesterol) are structural lipids that are also the building blocks for steroids, which have important metabolic functions.

Unlike proteins, lipids are remarkably durable; heating is not destructive but instead seems to be protective in a variety of archaeological contexts. Lipids are most often recovered from porous ceramic vessels that are used to store or cook food. Cooking pots are particularly valuable sources of lipids because boiling water-based liquids (soups, stews, braises) cause fats to float to the top of the liquid, where it can be readily absorbed into the porous ceramic fabric of the pot. This area on the interior of a pot can then be sampled for lipid analysis, allowing reconstruction of the foods cooked in the vessel. Analysis works by identifying the fatty acids present and their relative proportions, and then comparing those with known plant and animal lipids in a database, resulting in a match for the lipid (or lipids) that were used in the pot. Storage vessels, such as Roman amphorae used to ship olive oil around the Mediterranean Sea, can be analyzed using the same method.

Below are experimental modern pots used to calibrate methods for analyzing ancient food residues. Read about them in this short [New York Times](https://www.nytimes.com/2018/01/15/science/ancient-food-residues.html) (<https://www.nytimes.com/2018/01/15/science/ancient-food-residues.html>) article.



Roman amphorae, or liquid-storage vessels, at Pompeii. These ceramic vessels typically held oil, wine, or fish sauce and were sealed and shipped around the Mediterranean world.

Source: [Wikimedia Commons](https://commons.wikimedia.org/wiki/File:Ancient_Roman_amphorae_in_Pompeii.jpg)

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One concern with the archaeological study of lipids from cooking pots is the possibility of contamination. Even a single human fingerprint can impart skin oils that can be measured through the sensitive instruments used to detect lipids! For that reason, ceramics for lipid analysis are ideally excavated by archaeologists wearing gloves and are placed directly into sealed containers without cleaning. This careful protocol keeps lipids and other residues adhering to the surface of the ceramic in place.

Small Organic Molecules

A wide variety of additional organic compounds are characteristic of different foods and can be recovered from pottery vessels, in much the same way as lipids. Small organic molecules come in a wide variety of chemical configurations but can be analyzed with similar instruments. Some of these compounds are uniquely characteristic of specific classes of foods. One example is tartaric acid, which is the compound that forms cream of tartar (in a salt form) but is present naturally in high quantities in grapes and thus is a residue of wine. Other organic molecules include certain